Newton School of Technology × rishihood university

HACKTOBERFEST 26

Hack Day

12 hours. Open-source AI. Real problems.

31 OCT - 1 NOV

SONIPAT

NST · RISHIHOOD UNIVERSITY

MLH MAJOR LEAGUE HACKING · DEV · Gemma



OPENING ADDRESS

Abhishek Rathi

Director, Newton School of Technology

Rishihood University, Sonipat



TALK · MLH × NST × WEMAKEDEVS

Anuj

IIT Roorkee
Alumnus

7 years
In tech

NST
Faculty

System & AI essentials

The young folks' round table

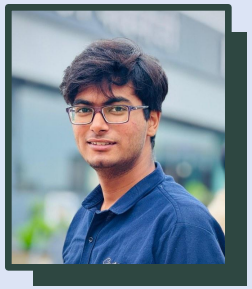
MODERATOR · NST FACULTY



Aman Agarwal

IIT Roorkee · 7 years in tech

Worked at **Flipkart, Adobe, Blinkit** and **Microsoft**



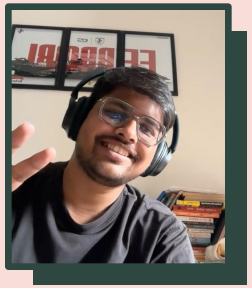
Sumit Goyal

- Kubernetes member
- LFX'26 mentor · CNCF



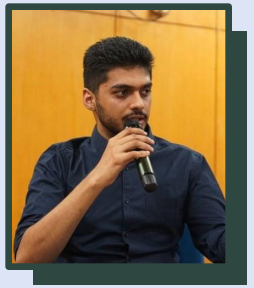
Paras

- GSoC'26 · Catrobat
- Samsung R&D intern



Abhijitam

- Full-stack engineer
- AI-native · Inovara



Aru Sharma

- GSoC mentor
- Summer of Bitcoin

TALK · ANUJ, NST FACULTY

MLH × NST × WeMakeDevs

MAJOR LEAGUE HACKING

Hackathons

The global student hacker community. Brings Hack Day to Sonipat for Hacktoberfest.

Newton School of Technology

Building

Where you learn by shipping real things, not only by passing exams.

Learning in public

A community that helps students start in open source and tech careers.

BEYOND ACADEMICS AT NST

- Clubs
- Hackathons
- Open source
- Industry projects

Session by WeMakeDevs

30 minutes



Kunal Kushwaha

FOUNDER



Aayush

COO



Sachin

CMO



HACKATHON BRIEFING

Sai Udayagiri

MSc
Computational Physics

2
Published research papers

GSoC
Contributor and mentor at SymPy

TU Delft
GSoC work published, Netherlands

HACKTOBERFEST HACK DAY · SONIPAT

Welcome. Let’s build.

BEFORE

Counting contributions. It led to spam.

TODAY

Solve a real maintainer problem with open models.

YOUR TIME

12 hours from idea to real.

NST · RISHIHOOD UNIVERSITY

The format

1-4

members per team. Solo is fine too.

12 hrs

6:30 PM to 6:30 AM

Food

Evening snacks, dinner, 2 AM refreshments, breakfast and lunch.
Your ID card is your token.

How it works

1

Pick an idea

One of 8 problems. Problem 8 is the Gemma track

2

Create a repo

New, empty, public, with an open-source license

3

Submit the repo link

By 8:30 PM

4

Init commit

HACKATHON.md: problem, models and why, team, members

5

Build through PRs

Every change is a PR with CI. Last merge by 6:30 AM

Start and end with a commit

FIRST · INIT COMMIT

Before you build, commit HACKATHON.md with:

- The problem statement you picked
- The models you're using, and why
- Your team name
- Every team member's name

LAST · END COMMIT

Your final push, by 6:30 AM.

We fetch your repo at 6:30 AM. That version is the one we score. Anything pushed later doesn't count.

Repo link by 8:30 PM

Build with open models

COUNTS · UP TO 15 POINTS

Open-weight models

Llama, Gemma, Qwen, Mistral, DeepSeek, Phi and similar.

Run them locally, or on our model server: [model list announced at the briefing].

SCORES 0 HERE

Closed model APIs

OpenAI, Claude, Gemini and other paid APIs.

Our test machines have no API keys, so these can't run in our automated tests either.

Name every model, and why, in HACKATHON.md. Claim one you don't use: 0 here and −5 overall.

Coding assistants (Copilot, Cursor, Claude Code) are fine. Name them in each PR.

We test your repo all night

Three things in your repo let us review it fairly:

`HACKATHON.md`

Your first commit. Team name, members, problem, and each model you use with the reason.

`CI that passes`

A GitHub Actions workflow that installs your project and runs your tests. It must pass on your last commit.

`examples/`

A sample input and the real output your tool produced from it. Proof that it works.

We check: CI passes, features are really built (not stubs), your tests, no secrets, and that the models you name are the ones your code uses.
Templates are in the Problem Statements doc.

Work like an open source project

1

Init commit, then main is PR-only

HACKATHON.md goes straight to main. After that, no direct pushes.

2

Every change is a PR with CI

A GitHub Actions workflow installs your project and runs your tests on every PR. Solo? Merge your own.

3

Every PR names its models

Models your product uses in this PR, and any AI coding assistant that helped write it.

Worth 10 points. Add .github/pull_request_template.md so every PR asks for its models. The template is in the Problem Statements doc.

Four rules

1. No earlier code or prototype. Start from an empty repo.
2. Public GitHub repo with an open-source license (MIT or Apache-2.0).
3. No API keys or secrets in your repo. Ever.
4. Use AI tools, but explain what you built. Or out.

Timeline

31 OCT

6:30 PM
Hackathon starts

7:30 – 9:30 PM
Dinner

8:30 PM
Repo link and init commit due

UNTIL 12 AM
Mentors clear your doubts

1 NOV

2:00 AM
Midnight refreshments

6:30 AM
Hackathon ends · repos frozen

8:30 AM
Presentations, 8 panels

1:30 PM
Results and prizes

Mentors and food

PROJECT DOUBTS · UNTIL 12 AM

Krishna Neetanshu

Kushagar Sai

Stuck? Ask early. Mentors help until midnight. After that, the build is yours.

FOOD · ID CARD = TOKEN

UNTIL 6:30 PM	Evening snacks
7:30–9:30 PM	Dinner
2:00 AM	Midnight refreshments
UNTIL 9:30 AM	Breakfast
12:30–2:30 PM	Lunch

No ID card, no food. Keep it on you all night.

PRESENTATIONS · 8:30 AM

Final evaluation

8 min

pitch per team

4 min

Q&A

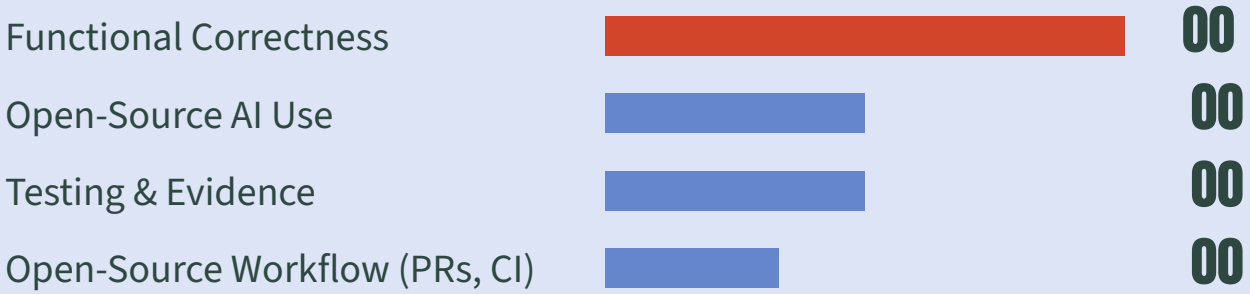
2 min

Buffer

Show: live demo · how it works · models and tools you used · what you learned. **Judges can ask any team member.**

How you're scored

00 AUTOMATED TESTED ON YOUR REPO



Claim open models you don't actually use: 0 for Open-Source AI Use and −5 overall.

00 PANEL YOUR PRESENTATION



Must pass: public repo with an open-source license, and any member can explain the code.

Submit on OrganizerHQ

1

Sign in, open your event

2

Challenges, then Add Submission

3

Enter repo, name, description, tech

4

Pick your challenges, Submit Project

Step-by-step guide, with how to add teammates

Challenges to pick: Best Open-Source AI Project (every team) · Best Use of Gemma 4 (Problem 8 teams only)

One submit, whole team

DEADLINE: 6:30 AM

Add teammates after:

Your Submission → Add Member

Submit your project

Cash prizes

2nd Prize

Rs 15,000

1st Prize

Rs 25,000

3rd Prize

Rs 10,000

Gemma Track

Rs 5,000

Gemma

TOP 3: PROBLEMS 1–7 · GEMMA TRACK: PROBLEM 8 ONLY

Tonight's blog challenge

2 × Rs 1,000

Write about Hacktoberfest and your project experience.

Submit your link by **[time]**.

One blog, two places: the same post counts on DEV and for us.

1 · POST ON DEV

dev.to/mlh-hackathon

2 · SEND US THE LINK

Our blog form

HACKTOBERFEST HACK DAY · SONIPAT

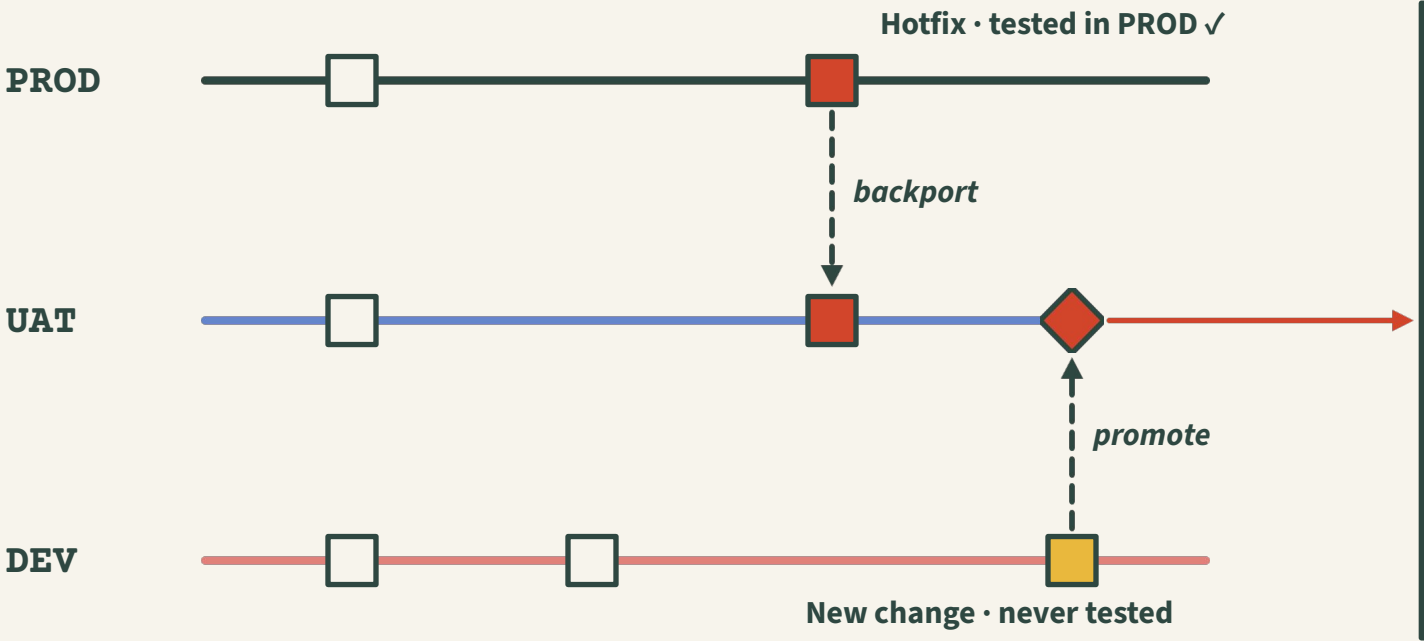
Problem Statements

7 MAINTAINER PROBLEMS · 1 GEMMA TRACK · DETAILS IN YOUR DOC

NST · RISHIHOOD UNIVERSITY

PROBLEM 1

Merges that know what's tested



CONFLICT · settings.py

Side A · hotfix #812
In PROD · CI passed ✓

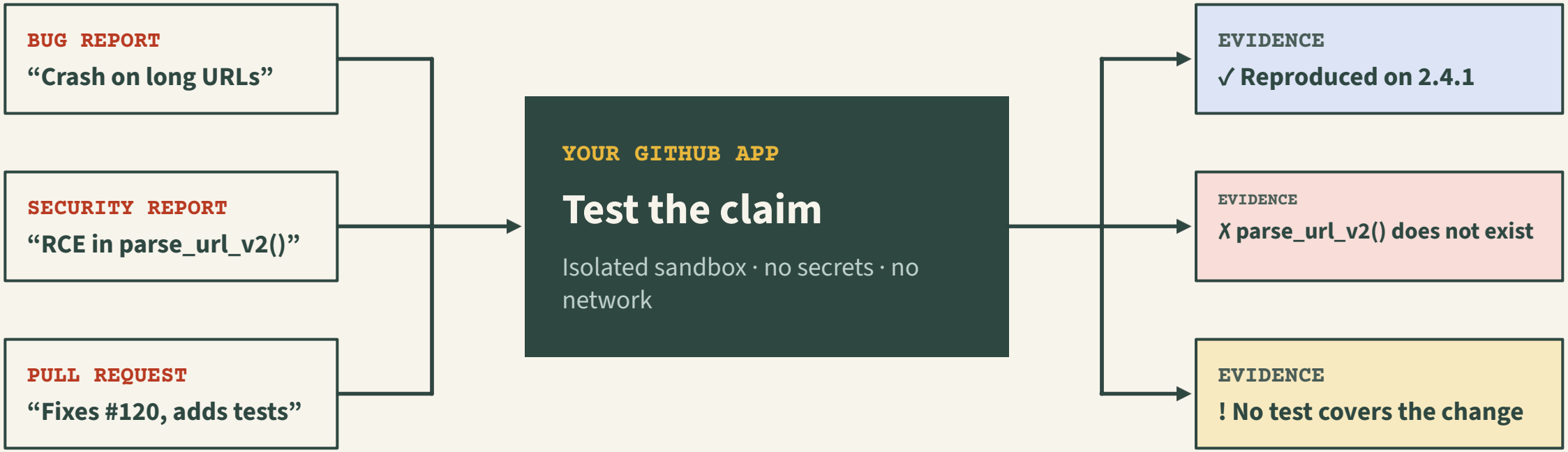
Side B · PR #845
DEV only · CI never ran

Then a normal Git merge commit.

You build: follow each change across branches, predict conflicts, and show which side was tested.

PROBLEM 2

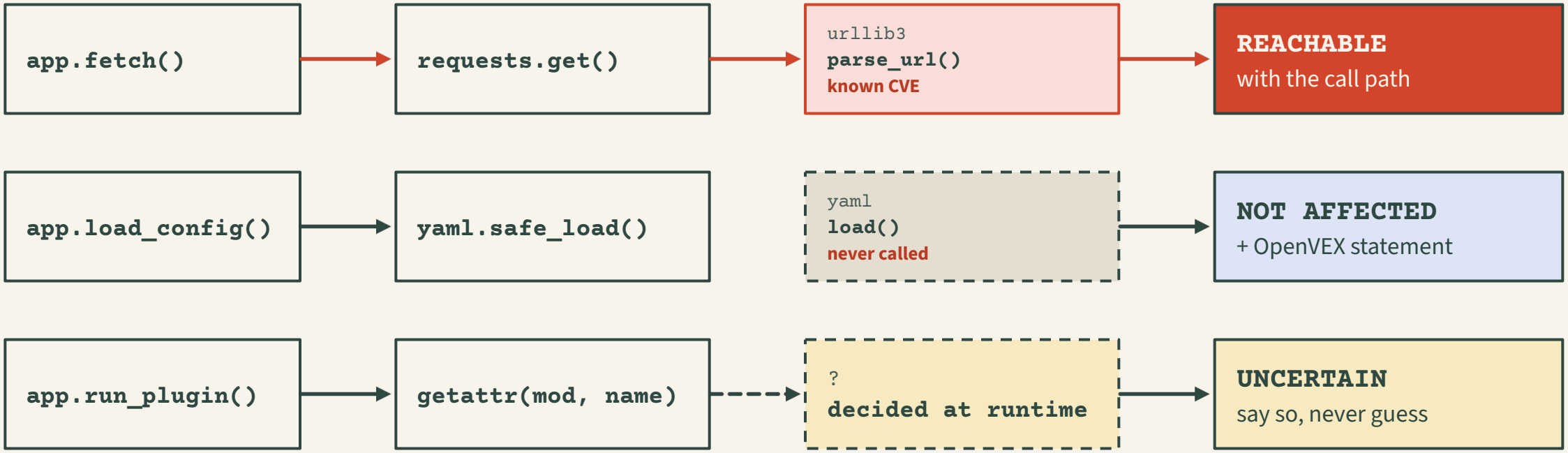
Triage that tests the claims



You build: a GitHub App that checks whether a report is true, safely, and shows maintainers the evidence.

PROBLEM 3

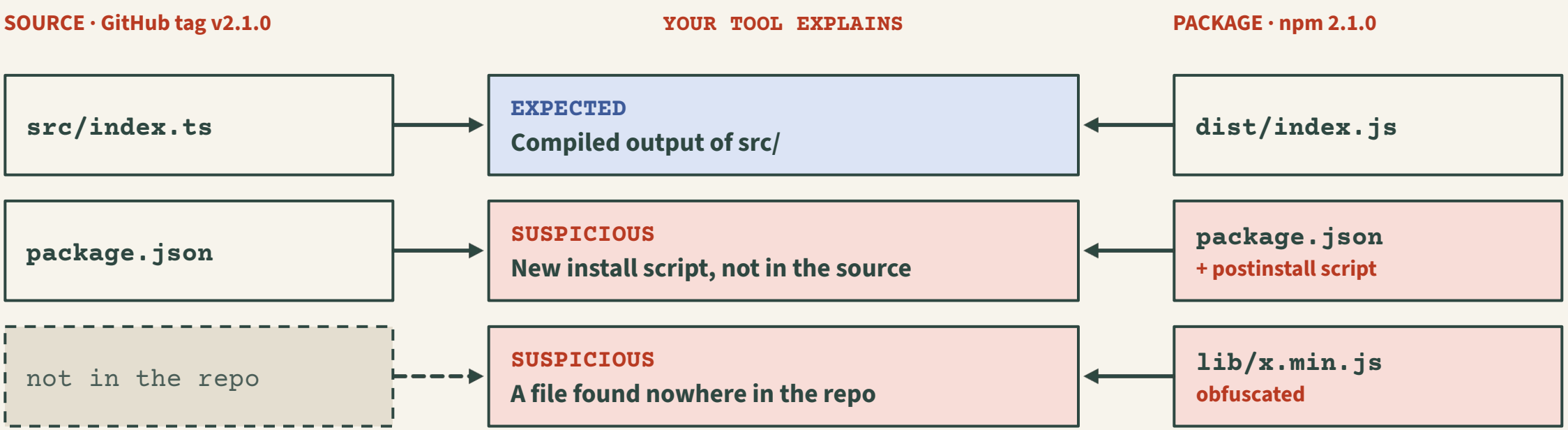
Which alerts can actually reach you?



You build: for each vulnerability alert, decide whether your code can reach the vulnerable function, with the call path as proof.

PROBLEM 4

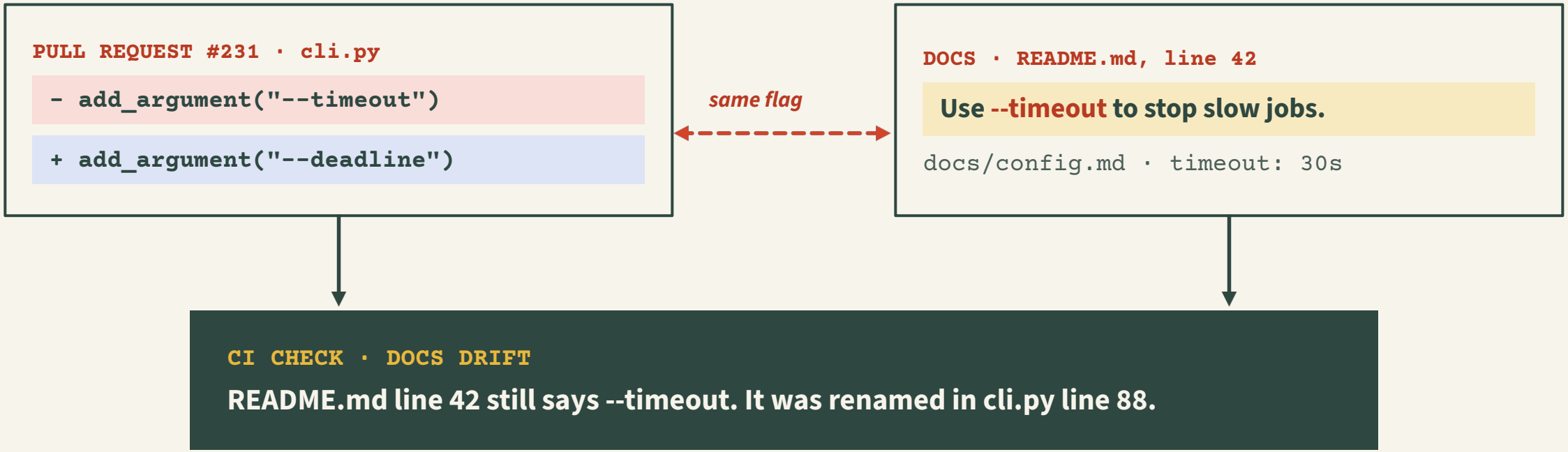
Does the package match its source?



You build: compare a published package with its source, explain every difference, and alert on suspicious releases.

PROBLEM 5

Catch docs a change made wrong



You build: a CI check that finds the exact doc sentence a pull request just made wrong.

PROBLEM 6

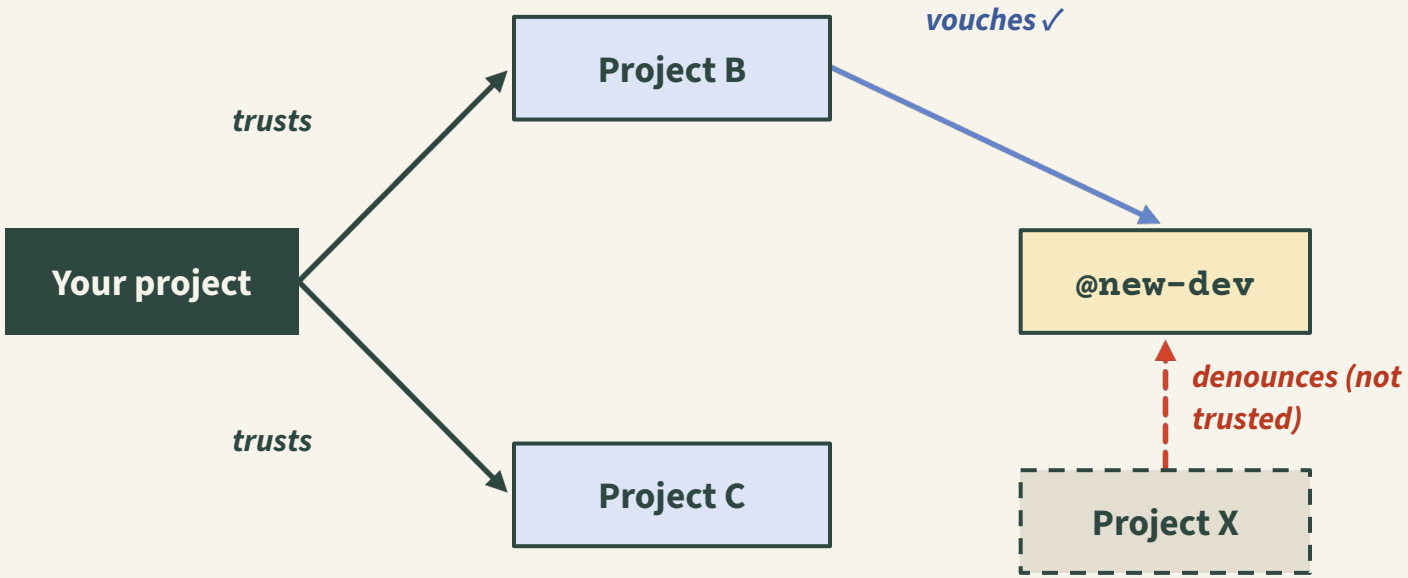
Recover the decisions nobody wrote down



You build: pull decisions and their reasons out of old discussions, and surface them when someone touches that code.

PROBLEM 7

A trust network against AI slop



NEW PR FROM @new-dev

Let in: vouched by Project B, which you trust.

X's denounce is ignored: you don't trust X.

CAMPAIGN DETECTED

38 near-identical PRs from 3-day-old accounts hit 40 projects in 2 hours.

Flagged, with the evidence.

You build: projects share whom they vouch for, choose whom they trust, and catch spam campaigns early.

PROBLEM 8 · GEMMA TRACK · RS 5,000

Build anything, with Gemma running on your laptop

Gemma

WHAT TO BUILD

Any original project where Gemma does the core AI work. It doesn't have to be one of Problems 1–7. Good directions:

- A multimodal tool that reads text and images
- A tool for learning, creativity or your community
- A tool that helps open source maintainers, offline

Try the Gemma demo first

THE ONE BIG RULE

Gemma runs on your own laptop. No cloud AI APIs at all.

Use Ollama, llama.cpp, LM Studio or transformers. Your app may use the internet for other things, like GitHub or data, but never for the AI.

Using any other model or cloud AI disqualifies the team.

Problem 8 teams compete only for the Gemma prize, not the top 3.

PROBLEM 8 · GEMMA TRACK · RULES AND JUDGING

Gemma only. On your laptop.

BEFORE YOU START

Download the model at break: it's several GB.

```
ollama pull gemma4:e4b
```

Low on RAM? gemma4:e2b. The 26B needs a 16 GB+ GPU.

MENTOR CHECKS

A mentor visits during the night to see Gemma running on your laptop.

Be ready to show ollama ps and your code calling localhost.

ZERO TOLERANCE

Any other model or any cloud AI API, even once, disqualifies the team.

Scoring: the same 100 points. Functional Correctness (30) here = it runs, and does what your examples show.

Repo rules still apply: HACKATHON.md, CI that passes, an examples/ folder, PRs with CI, open-source license.

Blog winners

OUR 15-DAY BLOG CHALLENGE

Congratulations to our 3 winners

1. Rajat Srivastav
2. Prachi Vats
3. Yatharth Khandelwal

TONIGHT · 2 × RS 1,000

The two best overnight blogs

1. Best Blog 1
2. Best Blog 2

Happy Hacktoberfest!

Share your feedback

Claim partner perks

.tech domain	Free for 1 year
Render	$50 credit
ElevenLabs	3 months of Creator
Tinker	$10 credit
Backboard	$5 credit

HACKTOBERFEST HACK DAY · SONIPAT

Thank You Volunteers for Making Hacktoberfest Hack Day a Massive Success!

devclub.



NST · RISHIHOOD UNIVERSITY